

Computing Teaching Reference Handbook

Programming foundations

Variables hold values that can change while a program runs. A beginner lesson should connect the variable name to its purpose, not just show assignment syntax. Useful examples include a score, temperature, username, or running total.

Conditions choose between paths. Learners often confuse assignment with comparison, or forget that only one branch may run in a simple if/elif/else chain. Short trace tables help make the control flow visible.

Loops repeat work. A for-loop is useful when iterating over a known collection or range. A while-loop is useful when repetition depends on a condition. A common beginner error is creating a while-loop whose condition never changes.

Functions group reusable logic. Learners should distinguish defining a function from calling it, parameters from arguments, and print from return. Examples should keep side effects small so the return value is easy to see.

Lists hold ordered values. Beginner work should cover indexing, iteration, adding/removing items, and the fact that index 0 refers to the first item.

Git and collaboration

A Git repository stores the project history. A branch gives a developer a place to work without changing the main branch immediately. Commits should represent understandable changes rather than one huge dump at the end of a week.

A merge conflict is not a broken repository. It means Git found overlapping edits and needs a human to decide the final content. Learners should read both sides of the conflict before editing the file.

A pull request is a review point. A useful PR description says what changed, how to test it, and anything still unfinished. Secrets such as API keys do not belong in commits.

Adding a file to .gitignore prevents future tracking when configured correctly, but it does not erase a secret that was already committed. Exposed keys should be rotated.

SQL basics

SELECT chooses columns or expressions to return. WHERE filters rows before grouping. JOIN combines rows from related tables. GROUP BY creates groups for aggregate functions such as COUNT, SUM, AVG, MIN, and MAX.

A query can be syntactically valid and still be logically wrong. Joining users.id to orders.id instead of orders.user_id may return believable data by accident. Examples should include this kind of mistake.

SQLite is useful for small local applications because it has no separate server process. It still uses relational tables and SQL, so the same core query ideas carry over to larger databases.

REST and FastAPI

An HTTP request has a method, URL, headers, and sometimes a body. A response has a status code, headers, and usually a body. The method should match the action being performed.

FastAPI maps routes to Python functions. Pydantic models describe the expected input shape. When a required field is missing, validation should fail before business logic runs.

Status codes are part of the API contract. A missing resource is different from a server crash, so 404 and 500 communicate different problems.

Practice ideas

Ask learners to trace a short program before they write one. Give a broken SQL JOIN and ask what data it would connect. Give two API requests and ask which one should be GET versus POST. Give a small Git history and ask where a secret leak occurred.

Good computing lessons should make internal state visible: variable values, loop iterations, request bodies, response JSON, SQL rows, and Git history.